Claire Kim

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Education

University of California, Los Angeles (UCLA) | Bachelor of Science in Mechanical Engineering ***Sep 2025 – Jun 2029***

Experience

Undergraduate Researcher **Sep 2026 - Present**
Micro and Nano Manufacturing Lab *Los Angeles, CA*

- Assembled and coated Electrowetting-on-Dielectric (EWOD) chips using precision fabrication techniques within the UCLA Nanofabrication cleanroom, optimizing chip processing to eliminate errors in oil-free operation
- Operated EWOD devices, analyzed CAD models and electronic hardware documentation to procure custom parts, and successfully assembled a new system replica from scratch based on technical specifications

Projects

Product Engineer **Jul 2026 - Sep 2026**
Matcha-matic | Independent Project *Camarillo, CA*

- Designed a compact, low-cost ($50 budget) automated matcha maker from scratch in SolidWorks, using FDM 3D printing and iterative design for portable functionality
- Engineered a magnetic stirrer powertrain using a brushed DC motor, dual side magnets, and custom aerating attachments to fully aerate and mix matcha and cold foam without clumping
- Integrated an ESP32 microcontroller, motor driver, and silicone-tube water distribution system to execute multi-mode automated programs via a 4-button user interface

Mechanical Engineering Lead **Apr 2026 - Jun 2026**
Petalux | Creative Labs *Los Angeles, CA*

- Engineered the mechanical structure of an automated kinetic lamp in Autodesk Fusion 360 from requirements to fully constrained CAD assemblies and prototypes
- Designed a custom powertrain with central screw linkage and specialized petal gears to actuate a 10-petal opening sequence while minimizing axial wobble through tight tolerances
- Developed space-efficient internal chassis routing for electronics and wiring without obstructing moving parts; collaborated with cross-functional teams to integrate UI hardware via snap-fits and custom cutouts

Mechanical Engineering Lead **Oct 2025 - Nov 2025**
Aeris | Theta Tau *Los Angeles, CA*

- Led the mechanical design of an innovative smart-cooling system integrating facial recognition tracking and tri-axis actuation to achieve dynamic, automated airflow control and directional cooling
- Engineered linkages and custom enclosures in SolidWorks and Fusion 360 using Design for Assembly (DFA) principles, streamlining Arduino hardware integration via high-precision friction fits
- Executed the full product lifecycle from FDM 3D printing to electromechanical validation, performing precision soldering and iterative troubleshooting to eliminate hardware errors

Chassis Mechanical Engineer **Sep 2025 - Jun 2026**
BB-8 | X1 Robotics - American Society of Mechanical Engineers *Los Angeles, CA*

- Designed and assembled BB-8 drive mechanisms in SolidWorks, optimizing weight distribution and omnidirectional motion stability
- Positioned motor housings and drivetrain components to fit within a strict 2mm machined shell tolerance while maximizing internal volume
- Modeled the core support ring in CAD, validating structural stress tolerances and in-house manufacturing feasibility for rapid fabrication

Technical Skills / Certifications

- **Modeling:** SolidWorks, Fusion 360, AutoCAD, FEA, DFM, Tolerance Analysis, Rapid Prototyping
- **Electronics & Manufacturing:** Arduino, Soldering, Machining, Welding, 3D Printing, CNC
- **Programming Languages:** Python, C++, JavaScript, CSS/HTML
- **Relevant Coursework:** Statics, Engineering Design/CAD, MATLAB, Fluids, Dynamics, Thermodynamics
- **Certifications:** Certified Solidworks Associate (CWSA), OSHA Safety